

Bug # 5 → To many API calls while searching

* Where :- useTasks.js (Frontend)

The API call was directly being made inside the `useEffect()`.

* How I found it

I opened the Network tab and tested the search bar by typing a word one character at a ~~time~~ time.

I noticed every change in the search input triggered `FetchTasks()`;

for eg:- abcdef

When typing a → server call

ab → server call

abc → server call

* Root cause

There was no delay between changing the search input and sending the API request.

∴ query in `useEffect` dependencies, immediately triggered API call.

This created unnecessary network traffic.

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★ fix :- I added a 300ms debounce using `setTimeout()` and wrapped the `fetchTasks` inside it

```
const timerId = setTimeout(() => {  
  fetchTasks(....)  
}, 300);
```

~~to~~ This ^{reset} 300ms timer for ever change in search character.

If user enters another character the new timer of 300ms starts

★ Result :- The API is no longer called for every keystroke.
It ~~wait~~ waits until the user pauses typing for about 300ms before sending the request.

eg :- a $\xrightarrow{\text{wait for 300 ms}}$ ab $\xrightarrow{\text{wait}}$ abc

↓ user take pause
Request sent to server

∴ abc → server
abcde → server